

# Autodesk® **Scaleform**®

## Scaleform

Scaleform 4.0

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## Autodesk® Scaleform® 4.4

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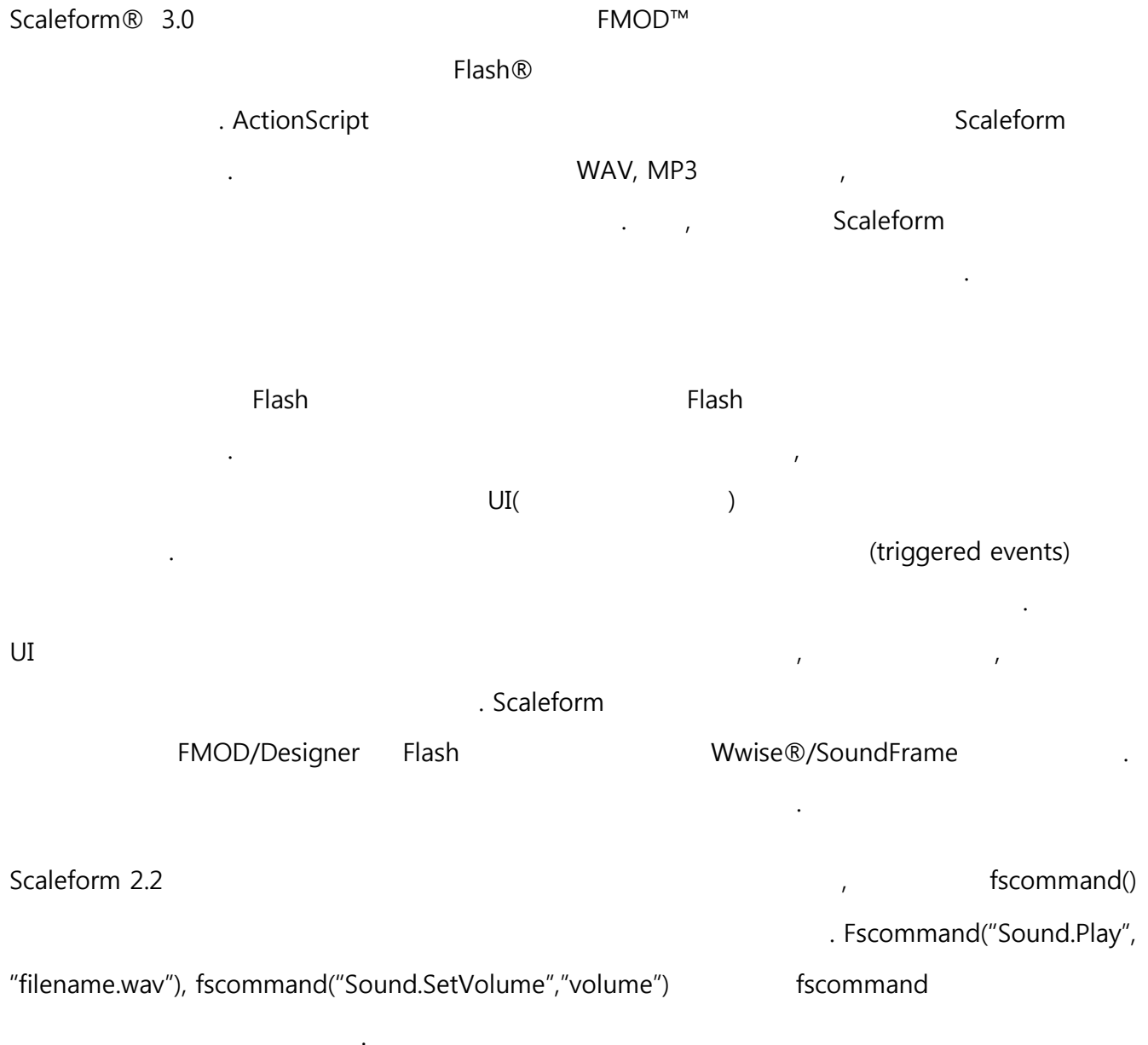
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# 1 소개



```

Scaleform                                Gfx::Audio                                Gfx::Loader
.
SWF
Sound::SoundRenderer                                . Sound::SoundRenderer
C++                                . Scaleform
FMOD                                .
:
SoundRenderer* pSoundRenderer = GetSoundRenderer();
if (pSoundRenderer)
{
    // Sound renderer and parameters for playing SWF streaming sounds
    // are provided to the loader by Gfx::Audio. These parameters
    // control when and how the synchronization between SWF frames
    // and sound frames should be done.
    Ptr<Gfx::Audio> pAudioState = *new Gfx::Audio(pSoundRenderer);
    loader.SetAudio(pAudioState);
    pSoundRenderer->SetMasterVolume(SoundVolume);
    pSoundRenderer->Mute(IsMute);
}

Flash
.
Wwise
.
Scaleform Player                                Flash

```

### 3 비디오 사운드 시스템 초기화

```

                                                                    Gfx::Video::VideoSoundSystem
                                                                    . Scaleform
Gfx::Video::Video
Sound::SoundRenderer
:
SoundRenderer* pSoundRenderer = GetSoundRenderer();
if (pSoundRenderer)
{
    // Setting a video sound system instance which is based on
    // SoundRenderer interface
    pVideo->SetSoundSystem(pSoundRenderer);
}

:
// Setting a video sound system instance which is specific to
// a particular platform
pVideo->SetSoundSystem(
    Ptr<Video::VideoSoundSystem>(*new Video::VideoSoundSystemDX8(0)));

SetSoundSystem()          Gfx::Video::Video
                                                                    .
```

## 4 사운드 이벤트 인터페이스

Scaleform 3.2 FMOD/Designer Flash  
Wwise®/SoundFrame(Sound::SoundEvent::PostEvent Sound::SoundEvent::SetParam)  
ActionScript  
Scaleform Flash player  
Scaleform

1. Sound::SoundEventFMOD - FMOD Ex API
2. Sound::SoundEventFMODDesigner - FMOD EventSystem API , FMOD Designer
3. Sound::SoundEventWwise - Wwise API
4. Sound::SoundEventWwiseSF - Wwise SoundFrame

:

```
#include "Sound/Sound_SoundEventWwise.h"  
Ptr<Sound::SoundEvent> soundEvent = *new Sound::SoundEventWwiseSF;  
  
soundEvent->PostEvent("Sound_PostEvent", "Play_Hello");  
soundEvent->SetParam("Sound_SetParam", "Enable_Effect", 85);
```

Scaleform Player , Flash



## 5 비디오 사운드 시스템 인터페이스

GFx::Video::VideoSoundSystem    GFx::Video::Video

Video::SetSoundSystem()

GFx::Video

- Video::VideoSoundSystemDX8    –    Windows    DirectSound
- Video::VideoSoundSystemXA2    –    Windows    Xbox360    XAudio2
- Video::VideoSoundSystemPS3    –    PS3    MultiStream
- Video::VideoSoundSystemWii    –    Wii
- Video::VideoSoundSystemFMOD - FMOD
- Video::VideoSoundSystemWwise – Wwise

Scaleform

Scaleform

Sound::SoundRenderer

GFx::Video::VideoSoundSystem

Video::VideoSoundSystem

Video::VideoSound

Sound::SoundRenderer

Sound::SoundRenderer

Video::Video

.(

.)

:

`#include "Video/Video_VideoSoundSystemXA2.h"`

`pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new VideoSoundSystemXA2(0, 0)));`

```

        Video::VideoSoundSystem    Video::VideoSound
        VideoSoundSystem
VideoSoundSystem                    VideoSound
        , Create                    . Scaleform
        . (                          .)    VideoSound
        , Scaleform
        VideoSound::PCMStream
        VideoSoundSystem
        .
        VideoSoundSystemWwise    Wwise SDK    v2009.2.1 build 3271
        . Audiokinetic
        /
        SDK\samples\Plugins\AkAudioInput    Wwise
        . Scaleform Video    Wwise SDK
        .
        :
        #include "Video/Video_VideoSoundSystemWwise.h"
        Ptr<Video::VideoSoundSystem> wwiseSound = Ptr<Video::VideoSoundSystem>(*new
            VideoSoundSystemWwise());
        pVideo->SetSoundSystem(wwiseSound);
        wwiseSound->Update();

        Scaleform Player
        . ,
        Getting Started with Video
    
```

## 6 예제 소스 코드

- Scaleform Player , FMOD/Wwise :
  - o Apps\Samples\FxPlayer\FxPlayer.cpp
  - o Apps\Samples\Common\FxSoundFMOD.cpp
  - o Apps\Samples\Common\FxSoundWwise.cpp
- FMOD :
  - o Src\Sound\Sound\_SoundRendererFMOD.h
  - o Src\Sound\Sound\_SoundRendererFMOD.cpp
- (FMOD/Designer Wwise/SoundFrame):
  - o Src\Sound\Sound\_SoundEventFMOD.h
  - o Src\Sound\Sound\_SoundEventWwise.h
- :
  - o Src\Video\Video\_VideoSoundSystem\*.cpp
- Flash :
  - o Bin\Data\AS2\Samples\Audio\_Demo\_AS2 fla
  - o Bin\Data\AS3\Samples\AudioDemo\_AS3 fla

## 7 추가 자료

CRI Movie ,  
(export)

ActionScript

SWF

- [Adobe Flash ActionScript 2.0](#)
- [Scaleform - Flash Support Overview](#)
- [Getting Started with Video](#)
- [CRI Movie Encoder Command Line Tool Overview](#)
- [GFxExport Reference Guide](#)